

Notation	Description
A, B, C	Capital letters represent matrices
$\mathbf{u}, \mathbf{v}, \mathbf{w}$	Lowercase letters represent vectors
$A \in \mathbb{R}^{m \times n}$	Matrix A has m rows and n columns
A^T	The transpose of matrix A
$\mathbf{v}^T$	The transpose of vector $\mathbf{v}$
A^{-1}	The inverse of matrix A
$\det(A)$	The determinant of matrix A
AB	Matrix multiplication of A and B
$\mathbf{u} \cdot \mathbf{v}; \langle \mathbf{u}, \mathbf{v} \rangle$	Dot product of vectors $\mathbf{u}$ and $\mathbf{v}$
$\mathbb{R}$	The set of real numbers (e.g., $0, -0.642, \sqrt{2}$)
$\mathbb{R}^2$	The set of 2D vectors, e.g., $\mathbf{v} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$
$\mathbb{R}^n$	The set of n -dimensional vectors
$\mathbf{v} \in \mathbb{R}^2$	Vector $\mathbf{v}$ is an element of $\mathbb{R}^2$
$\ \mathbf{v}\ _1$	L_1 -norm of a vector
$\ \mathbf{v}\ _2; \ \mathbf{v}\ $	L_2 -norm (Euclidean norm) of a vector
$T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$	Transformation T of $\mathbf{v} \in \mathbb{R}^2$ into $\mathbf{w} \in \mathbb{R}^3$

Table 1: Linear Algebra Notation Summary